

Smart Rider Safety & Hazard Mitigation System (SRSHMS) – Technical Progress Sheet

Track A (Embedded IoT System Development)
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Problem

Two-wheeler riders can suddenly encounter roadside hazards such as dogs or other moving objects, leaving very little time to react. A second, separate problem occurs after a crash: if the rider is injured or unable to use a phone, contacting someone for help becomes difficult. The project addresses both the prevention of hazard-related accidents and the ability to get help quickly if a crash does occur.

Proposed Solution

SRSHMS is an ESP32-based safety system built around a prevent → protect → rescue concept. A radar sensor detects nearby objects or movement, while a microphone with audio classification analyzes the surrounding sound (dog bark, human voice, vehicle/traffic sound, background noise). The radar and audio results are fused to estimate a possible dog-related hazard, and the rider is warned with directional left/right vibration motors, LEDs and an OLED display. When a dog-chase hazard is confirmed, the system also triggers a short, human-inaudible ultrasonic deterrent burst (20–25 kHz) aimed toward the detected direction, intended to move the dog safely away from the rider without any physical harm; the burst is brief, direction-limited and auto-cuts off once the hazard clears, so the effect on other nearby animals is kept to a minimum. The system also includes crash detection with a cancellable countdown; if not cancelled, it activates a local siren and sends an emergency notification with GPS location over Wi-Fi/IoT, while still providing local alerts if the network is unavailable.

System Architecture

Single ESP32-based controller with four layers: sensing, decision/control, rider alerting and emergency response.

- Sensing layer: radar/proximity sensor, microphone (audio classification), MPU6050 (crash/IMU), GPS module.
- Processing layer: ESP32 – sensor processing and filtering, hazard decision logic (sensor fusion of radar + audio), alert control, and communication.
- Rider alert layer: left/right vibration motors (via BJT driver circuits), LEDs, OLED status display.
- Deterrent layer: harm-free ultrasonic transducer (20–25 kHz, human-inaudible) that emits a short, direction-aimed burst to move the dog away — no physical contact or injury, and the burst auto-stops once the hazard distance clears.
- Emergency response layer: local buzzer/siren, crash countdown with cancel option, GPS-based IoT/cloud emergency notification.
- Key design point: radar alone does not identify a dog – it only detects a nearby object/movement; the audio classifier adds the “dog bark” signal, and the hazard decision is taken only when both agree.

Current Progress

- System architecture and main control/decision logic fully defined (radar + audio fusion → hazard decision → rider warning; separate IMU → crash → countdown → alert path).
- Proteus simulation in development to demonstrate sensor inputs, control logic and actuator circuits using controllable inputs in place of exact radar/microphone hardware models.
- BJT-based driver circuit designed for the left/right vibration motors.
- Core test scenarios identified (T01–T09) covering safe state, directional warnings, dog-hazard classification, crash countdown/cancel, and offline/local operation.
- Harm-free ultrasonic deterrent logic defined: fires only on a confirmed hazard (radar + audio agreement), as a short direction-aimed burst with automatic cutoff — no physical harm to the dog or unintended effect on other animals nearby.

Technology Stack

Hardware: ESP32, radar sensor, microphone, MPU6050, GPS module, OLED display, vibration motors, LEDs, BJT transistor drivers, ultrasonic deterrent transducer (20–25 kHz), buzzer/siren.

Software: ESP32 firmware, Proteus simulation, an audio ML/TinyML approach for sound classification, version-controlled on GitHub.

Communication: Wi-Fi and IoT/cloud messaging for emergency notification.

Testing

Nine scenarios are being validated in simulation, from no-hazard/safe state through left/right object warnings, dog-bark-plus-proximity classification (and correctly rejecting human voice or vehicle sound as non-dog), crash-triggered emergency countdown and cancellation, and continued local safety operation when the network is unavailable. Actual PASS/FAIL results will be recorded as hardware testing progresses.

Limitations

- Exact radar and microphone modules may not have matching Proteus simulation models, so controllable inputs currently stand in for real sensor outputs.
- Radar alone cannot identify a dog; hazard classification depends on the audio model, which outdoor noise can affect – this sensor-fusion reliability is our main technical risk.
- Sensor detection range may limit how early a hazard is identified, and cloud emergency notification depends on network connectivity.
- The current system is a simulation-based proof-of-concept and still requires physical hardware integration and field testing.
- The ultrasonic deterrent's directional beam is not perfectly narrow, so other animals within range could be mildly affected; beam width and placement still need physical tuning.

Next Steps

- Integrate the actual radar and microphone hardware and re-validate the fusion logic against real sensor data.
- Improve the audio classification model and carry out outdoor field testing.
- Refine sensor-fusion and crash-detection reliability to reduce false alarms.
- Strengthen GPS/IoT communication, build a compact wearable/vehicle-mounted prototype, and bench-test the ultrasonic deterrent's range/beam angle to confirm it reliably reaches the target dog with minimal effect on other animals nearby.